Bitter Lesson - Mid Submission Report
Legal SME for Civil Law: Contracts and Agreements (India)

Team: bitter lesson
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Executive Summary

This document reports the mid submission status of our project focused on a contract and agreement RAG-based Subject Matter Expert system intended for junior law associates. Scope covers the Indian Contract Act 1872 as the core statute but extends to a broad corpus of related material, including government General Conditions of Contract (GCC) documents, Model Concession Agreements (MCAs), tender templates, and relevant judgments and orders sourced from Supreme Court and High Court archives, all restricted to publicly available open sources. We have built a working pipeline: data acquisition, OCR to Markdown, preprocessing and chunking, dual index builds, and a qualitative retrieval evaluation across BM25, MPNet, and Legal-BERT. All scripts are open and reproducible.

Repository Layout

Data Acquisition

This section summarizes both our initial efforts and the more sophisticated scraping developed by a teammate. All sources are public. We retained authoritative PDFs and logged provenance for reproducibility.

Phase 1 - Initial collection via `wget`

Goal: small seed corpus of government documents and GCC templates for contract language reference.
Examples used:

- `https://www.iricen.gov.in/iricen/Works_Manuals/GCC-2022-ACS6.pdf`
- `https://dfccil.com/upload/GCC_2022_4DXM.pdf`
- `https://shipmin.gov.in/.../Model%20Concession%20Agreement_2021_...pdf`
- `https://www.pngrb.gov.in/pdf/public-notice/DGTA24062021.pdf`
- `https://gailonline.com/pdf/gcc/GCCCONSULTANCYSERVICES.pdf`
- Additional CPSE GCC and tender PDFs from BHEL, GAIL, IOCL.

Phase 2 - Manual case downloads from `sci.gov.in`

The Supreme Court site does not expose a convenient API. We curated a verified subset of judgments by manual search and download. Filenames were normalized by citation or date for traceability.

Phase 3 - Automated crawling of Indian Kanoon

We implemented a two stage crawler for `indiankanoon.org` to broaden coverage.

Scraping strategy

- **Query design** combined statute anchors and context terms, for example: "Section 10 Indian Contract Act validity", "Article 299 government contract", "power purchase agreement tariff CERC".
- **Session handling** with realistic headers to avoid light responses and handle pagination.
- **Link extraction** captured canonical `/doc/<id>/` links with deduplication.
- **Politeness** configurable delay between requests.

Downloader logic

- Reads discovered links, follows the PDF anchor, saves into `pdfs/`.
- Uses `downloaded.txt` and `failures.txt` for resumable runs.
- Filenames include the document id and judgment title.

Logging and resumption Append only log files enable safe resume on failure and clean deduplication.

Phase 4 - Corpus assembly

We assemble a clean corpus with logs for provenance and deduplication. This feeds downstream OCR and chunking. Planned next steps include richer metadata extraction, OCR fallback for scans, and incremental refreshes.

OCR to Markdown

We adopted DeepSeek-OCR with vLLM to batch page level OCR directly to Markdown. Each page is an image request in a batch.

Script

`batch_vllm_ocr.py`

Key features

- Inputs default to `../data` when run inside the `ocr/` venv.
- Outputs: per page `page.mmd` under `data_md/work`, merged `.md` under `data_md/md`, and a record in `corpus.jsonl`.
- Skip existing files using raw file sha256 and `corpus.jsonl` lookup to avoid re OCR.
- Per PDF progress via `tqdm` plus vLLM internal request bars.
- Provenance markers in merged files: `«<PAGE n»>`.

Notes on batching

On an L40S GPU, 32 pages per batch are stable at max tokens near 8192. Adjust as needed. You can autotune batch size by probing a warm up batch and backing off on out of memory.

Preprocessing and Chunking

Cleaning

DeepSeek grounding tags are stripped and common OCR noise is removed before chunking.

- Remove `<|ref|>...</|ref|>` and `<|det|>...</|det|>` tags.
- Drop empty table cells and runs of `<td></td>` like patterns.
- Collapse noisy punctuation such as ". ." into a single period and space.
- Normalize whitespace and dashes. Preserve statute numbering.

Section extraction

We detect sections with robust patterns for legal texts that may not use Markdown headings.

- Lines like "Section 74 ..." or "Sec. 74 ...".
- Numbered headings like "74. Title ...".

If no sections are found the entire document is treated as one section.

Chunking policy

We window by sentences into multi granularity chunks at 2048, 512, and 128 token targets. Chunk ids are deterministic and parseable:

```
<doc_id>::sec/<section_id>::g=<granularity>::o=<offset>
```

Neighbor pointers connect adjacent sections to support multi step retrieval. We store cleaned display text and a lowercased normalized text for indexing.

Indexing

Sparse

BM25 with rank_bm25. We index the 512 token granularity by default.

Dense

FAISS IndexFlatIP with cosine equivalence. We built two dense indexes:

- MPNet: `sentence-transformers/all-mpnet-base-v2`
- Legal-BERT baseline: `nlpaueb/legal-bert-base-uncased`

Evaluation Setup

We focus on a qualitative mid submission pass. We built 10 targeted queries and a batch evaluation script that prints a markdown table for side by side viewing.

Queries

Stored at `queries.jsonl`. Themes include offer and acceptance, S73 to S75 damages and penalties, S56 frustration, e contracts, arbitration agreement formation through electronic communications, and performance security in government templates.

Tools

- `batch_eval_to_md.py` - runs all 10 queries and writes a markdown table.

Pooling note for Legal-BERT

Legal-BERT is a masked language model. Cosine on CLS can be weak. For the mid pass we accept it as a baseline and may switch to mean pooling for a modest lift.

Result table

Full table can be found in ‘evals/‘

Qualitative Evaluation Summary The evaluation compared three retrieval backends — BM25 (sparse lexical), MPNet (dense semantic), and Legal-BERT (domain-tuned masked LM used as dense baseline) — across ten structured queries derived from statutory and contractual themes under Indian law.

- **BM25:** Excelled at retrieving verbatim clause-level text where keywords matched directly (e.g., “WhatsApp contract”, “Section 56 frustration”), and surfaced judgment facts and party communications accurately. However, its results were often repetitive across top chunks and lacked conceptual grouping for statute sections.

Question
What amounts to a valid offer and acceptance under Indian contract law, and how are offers through email or W
What amounts to a valid offer and acceptance under Indian contract law, and how are offers through email or W
What amounts to a valid offer and acceptance under Indian contract law, and how are offers through email or W
When can a penalty or liquidated damages clause be reduced by courts under Section 74 of the Indian Contract
When can a penalty or liquidated damages clause be reduced by courts under Section 74 of the Indian Contract
When can a penalty or liquidated damages clause be reduced by courts under Section 74 of the Indian Contract
Explain frustration under Section 56 and when a contract becomes void due to impossibility.
Explain frustration under Section 56 and when a contract becomes void due to impossibility.
Explain frustration under Section 56 and when a contract becomes void due to impossibility.
What is the difference between liquidated damages and penalty in Indian law, and which Supreme Court cases c
What is the difference between liquidated damages and penalty in Indian law, and which Supreme Court cases c
What is the difference between liquidated damages and penalty in Indian law, and which Supreme Court cases c
Are e-contracts and clickwrap agreements enforceable in India? Point to the statutory basis and key cases.
Are e-contracts and clickwrap agreements enforceable in India? Point to the statutory basis and key cases.
Are e-contracts and clickwrap agreements enforceable in India? Point to the statutory basis and key cases.
What are the usual termination for convenience and force majeure clauses in Government EPC/GCC templates
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What are the usual termination for convenience and force majeure clauses in Government EPC/GCC templates
Under Sections 73–75, how is compensation for breach assessed and what is the role of mitigation?
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How is an arbitration clause communicated via electronic correspondence treated at the prima facie stage under
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How is an arbitration clause communicated via electronic correspondence treated at the prima facie stage under
What are standard security/performance guarantee requirements in model concession agreements or EPC contra
What are standard security/performance guarantee requirements in model concession agreements or EPC contra
What are standard security/performance guarantee requirements in model concession agreements or EPC contra
What limitations exist on specific performance compared to damages, and what do Law Commission reports say
What limitations exist on specific performance compared to damages, and what do Law Commission reports say
What limitations exist on specific performance compared to damages, and what do Law Commission reports say

- **MPNet:** Produced the most contextually correct results overall. It consistently aligned queries with relevant statutory extracts (e.g., Sections 73–75, Section 74) and doctrinal explanations (e.g., *Fateh Chand v. Balkishan Das*). It captured semantic proximity even when exact legal terms varied, showing strong generalization to Indian legal phrasing despite not being domain-finetuned.
- **Legal-BERT:** Underperformed on sentence-level retrieval because CLS pooling from a masked LM is not optimized for cosine similarity. It frequently returned unrelated snippets (e.g., random contract templates or Specific Relief Act fragments). Mean pooling or fine-tuning on contrastive objectives is expected to improve future results.
- **Hybrid insight:** In most cases, a simple rank fusion (RRF) between BM25 and MPNet improved diversity while maintaining relevance, suggesting a viable hybrid baseline for the final system.

Overall, MPNet delivered the most legally coherent and interpretable retrievals, while BM25

served as a reliable factual complement. Legal-BERT remains a valid baseline for demonstrating domain adaptation effects but requires further embedding calibration. The evaluation confirms that dense retrieval with general-purpose sentence embeddings is already competitive for Indian legal text when combined with minimal preprocessing and document-level chunking.

Reproducibility - Commands

OCR

```
python batch_vllm_ocr.py \
  --input ../data --out ../data_md --tmp ../data_tmp \
  --dpi 300 --batch_pages 16 \
  --model_id deepseek-ai/DeepSeek-OCR \
  --prompt "<image>\n<|grounding|>Convert the document to markdown."
```

Chunk and index

```
python chunk_and_index.py \
  --md_dir data_md/md --out_dir data_out \
  --granularities 2048,512,128 \
  --bm25_granularity 512 --faiss_granularity 512 \
  --vector_model sentence-transformers/all-mpnet-base-v2 --device cpu

python chunk_and_index.py \
  --md_dir data_md/md --out_dir data_out \
  --bm25_granularity 512 --faiss_granularity 512 \
  --vector_model nlpaueb/legal-bert-base-uncased --device cpu
```

Batch evaluation

```
python batch_eval_to_md.py \
  --queries eval/queries.jsonl --chunks data_out/chunks.jsonl \
  --bm25 data_out/index/bm25/bm25_g512.pkl \
  --mpnet_index data_out/index/vector/faiss_sentence-transformers-all-mpnet-base-v2_g512.index
  --mpnet_ids   data_out/index/vector/faiss_sentence-transformers-all-mpnet-base-v2_g512.ids \
  --legal_index data_out/index/vector/faiss_nlpaueb-legal-bert-base-uncased_g512.index \
  --legal_ids   data_out/index/vector/faiss_nlpaueb-legal-bert-base-uncased_g512.ids \
  --device cpu --topk 3 --out_md eval/results.md --snippet_chars 300
```

Planned Work for Final

- Expand corpus with more GCC and sectoral MCAs and richer metadata extraction.
- Add a reranker to get better retrieval results.
- Add hybrid scoring with RRF lines in batch reports and a light reranker for top 50.
- Implement per page bounding box capture for highlight rendering and citation pointers.